ESTIMATOR-UPGRADE closure consolidation: the enumerated Reading-H selection margins are controlled-error, continuum, single- and two-shell

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The ESTIMATOR-UPGRADE gate asked that the Reading-H selection $\Delta F[\mathcal{R}] = F[\mathcal{R}] - F[\mathcal{R}_H] > 0$ – established at ESTIMATOR grade (point estimates from the fast pipeline, no certified error) – be upgraded to a CONTROLLED-ERROR bound. This note consolidates the five-note chain that delivers it and records the gate closure. It introduces no new computation; every number below is from a cited note's executed script.

2. Single-shell enumerated readings (LAM/HEX/FCC/BCC, $\mu^2 = 0.005$)

M-quadrature knob (estimator-upgrade-enumerated v1.0): the curvature $\kappa_{\mathcal{R}} = \Delta F''_{\mathcal{R}}(0) = 2\Delta F_{\mathcal{R}}(h)/h^2 > 0$ is certified at two loop-quadrature resolutions ($N_{\rm pt} = 6000, 20000$) with envelope $< 0.1\%$; $A = 0$ is a strict minimum and no reading condenses. Binding $\kappa_{\rm LAM} = 0.851$.

dI-quadrature + amplitude-grid knobs (estimator-upgrade-knobs v1.0): $\kappa_{\mathcal{R}}$ moves $< 0.1\%$ under `dI`-grid refinement $(6000, 50) \to (12000, 100)$; the no-condensate verdict is grid-monotone at $N_G = 121/241/481$.

Continuum (node-free) (estimator-upgrade-knobs v1.0): the curvature-chord bound $\Delta F(A) \geq \min(v_i, v_{i+1}) - \frac{1}{8} M_i \delta^2 > 0$ holds on every $A > 0$ interval (LAM $+8.8 \times 10^{-7}$, BCC $+2.5 \times 10^{-7}$): no inter-node condensate. The single-shell margins are thus controlled against all three numerical knobs and the no-condensate statement is continuum.

3. Two-shell $\{110\} + \{200\}$ ensemble (B1 point $r_{\rm bare} = 0.219$)

$(0, 0)$ **Hessian** (estimator-upgrade-knobs v1.0 + twoshell-continuum-bound v1.0): diagonal ($\kappa_{12} = 0$ by shell orthogonality; the m_{31} coupling is quartic), positive-definite, $\kappa_{\{110\}} = 5.16$, $\kappa_{\{200\}} = 3.86$. The softer direction is $\{200\}$ (a correction: the kernel penalty $Cq_0^4 > 0$ does not order the dressed curvatures).

Diagonal continuum (twoshell-continuum-bound v1.0): at $r = 0.219$ the M-minimised diagonal surface has $\min_{(A_1,A_2)\neq(0,0)} \Delta F_{\rm diag} = +3.9 \times 10^{-5} > 0$ and a 2D curvature-chord lower bound $+1.2 \times 10^{-4} > 0$. (This note also corrected the operating point: the legacy Math432 evidence runs at $r = 0.005$, not the B1 point.)

Exact-Wick anchored (twoshell-anchored-bracket v1.0 + twoshell-anchored-continuum v1.0): with the exact slogdet engine (Math432 reused by neuter-import, validated to 5×10^{-8}), the anchored free energy $\Delta F_{\text{anchored}} = \Delta F_{\text{diag}} + (F_{\text{exact}} - F_{\text{diag-basis}})$ has grid min $+4.6\times10^{-4} > 0$, a 2D curvature-chord bulk continuum lower bound $+1.3\times10^{-3} > 0$, and an $O(A^4)$ bracket near the origin so the anchored $(0,0)$ Hessian equals the diagonal one (PD). The exact-Wick no-condensate at the B1 point is therefore CONTINUUM on the whole (A_1, A_2) domain.

4. Closure

The estimator-grade Reading-H selection is upgraded to a controlled-error, node-free no-condensate statement, single- and two-shell. ESTIMATOR-UPGRADE is CLOSED at CONTROLLED-ERROR / STRONG-EVIDENCE grade (operator-authorized 2026-06-07). B1-RH-ENUM remains T6 CONDITIONAL on {H-LAYER, H-ADM-COH, SC-SCOPE}: the gate governed the MARGINS' error grade, not the selection SIGN, so the tier is unchanged.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) *"Is this a T7 theorem?"* NO, and the closure does not claim it: the curvature-chord lower bounds use discrete second-difference |Hessian| estimates (grid-stable, but not interval arithmetic), and the two-shell M-minimisation is over a finite grid. The grade is STRONG-EVIDENCE / controlled-error, which is exactly what the gate ("controlled error bound") requested. A T7 upgrade (rigorous |Hessian| upper bounds) is a separate, optional future step.

(β) *"Does closing the gate change B1's tier?"* DISMISSED: B1 T6 is the selection SIGN modulo three named hypotheses; ESTIMATOR-UPGRADE was the MARGINS' error-grade gate (GAP-2), never one of the three. Closing it removes a quality gate, not a hypothesis; the tier is unchanged.

(γ) *"Is the two-shell scope honest?"* The two-shell result is at the B1 point $r = 0.219$ over the Math432 scan domain; it is the EXACT-Wick anchored free energy (diagonal + bracket), continuum (bulk curvature-chord + near-origin PD). The named-but-not-pursued items (a T7 interval-arithmetic upgrade; a wider (A_1, A_2) domain) do not affect the closure at the stated grade.

Quantitative sanity check (mandatory): magnitude – all margins positive across five notes ($\kappa \geq 0.85$ single-shell; anchored continuum $+1.3\times10^{-3}$ two-shell); sign-direction – every curvature > 0 and every continuum lower bound > 0 (selection-preserving); limit-case – $\Delta F(0) = 0$ exactly and the engine reproduces Math432 to 5×10^{-8}; consistency – the anchored $(0,0)$ eigenvalues 5.16/3.86 are unchanged from the diagonal Hessian (bracket $O(A^4)$).

6. Result footer

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Result ID:             B1-RH-ENUM / ESTIMATOR-UPGRADE CLOSED (consolidation)
Precise statement:     The estimator-grade Reading-H selection dF>0 is upgraded
                       to a controlled-error, node-free (continuum) no-condensate
                       statement: single-shell (M, dI, amplitude-grid knobs +
                       curvature-chord continuum) and two-shell {110}+{200}
                       (exact-Wick anchored, continuum at r=0.219: bulk
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	curvature-chord +1.3e-3 + near-origin PD). Grade = CONTROLLED-ERROR / STRONG-EVIDENCE.
Scope:	Enumerated single-shell readings + two-shell ensemble at the B1 point; controlled error w.r.t. the named knobs; no-condensate continuum (strong-evidence, not interval arithmetic).
Dependencies:	estimator-upgrade-enumerated v1.0; estimator-upgrade-knobs v1.0; twoshell-continuum-bound v1.0; twoshell-anchored-bracket v1.0; twoshell-anchored-continuum v1.0.
Evidence grade:	EXECUTED (5 scripts) + STRONG-EVIDENCE (curvature-chord).
Reproduction command:	python codes/vacuum/estimator_upgrade_enumerated.py && python codes/vacuum/estimator_upgrade_knobs.py && python codes/vacuum/twoshell_continuum_bound.py && python codes/vacuum/twoshell_anchored_bracket.py && python codes/vacuum/twoshell_anchored_continuum.py
Expected output:	7/7, 13/13, 13/13, 7/7, 7/7 PASS.
Falsification gate:	A reading/amplitude with a negative continuum lower bound; a two-shell anchored surface point with dF<0 at r=0.219; a non-PD anchored (0,0) Hessian.
Tier before / after:	ESTIMATOR-UPGRADE: OPEN -> CLOSED@CONTROLLED-ERROR. B1-RH-ENUM: T6 CONDITIONAL -> T6 CONDITIONAL (unchanged; margins gate only). Operator-authorized 2026-06-07.
No-overclaim statement:	CONTROLLED-ERROR / STRONG-EVIDENCE, not a T7 interval-arithmetic theorem. The closure upgrades the MARGINS' error grade; the B1 T6 selection SIGN and its three named hypotheses are unaffected. SC-SCOPE / H-ADM-COH / H-LAYER remain B1's named hypotheses; G3PB-III stays open.
Next required action:	None for ESTIMATOR-UPGRADE (closed). Optional: a T7 interval-arithmetic upgrade of the curvature-chord bounds.